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Maximum Likelihood Pitch Estimation

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Abstract—A method for estimating the pitch period of voiced speech sounds is developed based on a maximum likelihood (ML) formulation. It is capable of resolution finer than one sampling period and is shown to perform better in the presence of noise than the cepstrum method.

I. INTRODUCTION

MANY current speech encoding techniques attempt to achieve low-rate digitized speech transmission by modeling the speech source as a linear filter, representing the vocal tract resonances, excited by either random noise or a quasiperiodic pulse train, representing the source signal for unvoiced and voiced speech, respectively. Achieving good-quality resynthesized speech with this model requires that both the filter parameters and the excitation signal be accurately estimated. The mechanical character of the early vocoders was due largely to their inability to extract the excitation signal and motivated the development of the voice-excited vocoder [1]. It is this same difficulty in accurately modeling the excitation signal which is responsible for the popularity of systems which transmit a quantized version of the estimated excitation signal even though they have a higher data rate than those which parameterize the excitation.

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The difficulty in estimating the excitation is compounded by departures from the idealizations used in developing the method which are encountered in a realistic situation: noisy environment, absence of the fundamental due to band-limiting, simultaneous presence of periodic and random excitation, phase distortion, or rapid changes in pitch period. In particular, the sensitivity of the pitch detector to ambient noise is a serious limitation in many potential applications of analysis-synthesis telephony systems [2].

The pitch detection scheme to be discussed is designed to be resistant to white, Gaussian noise, may be extended to colored noise, and shows promising performance in the presence of realistic environmental noise. In addition, it is capable of determining the pitch period with a resolution finer than one sample period, resulting in improved performance for high-pitched speech.

In revising this paper, the authors discovered that Noll [3] proposed a maximum likelihood (ML) pitch estimation method similar to that described in Section II of this paper. The modifications developed in Section III reduce his problem of multiple peaks with increasing amplitude caused by noise, and enable this method to provide a parameter which is useful in making the voiced-unvoiced decision. An interpretation in terms of the signal autocorrelation function is presented which is useful in suggesting possible efficient implementations and in providing insight into the frequency-domain behavior of the estimator. This method has been evaluated on several utterances with various types of noise and signal-to-

noise ratios and the results presented. These results indicate that this method is a potentially viable one for use in noisy environments.

II. ML FORMULATION

The problem is formulated as that of estimating an unknown periodic signal in white Gaussian noise of unknown intensity. Caprio has considered this problem in a previous paper [4] and the method presented here is a generalization of his result. The procedure will be developed in a discrete-time context.

Let s_k be a periodic repetition of the length P sequence q_k , i.e., $s_k = q_{k \bmod P}$. The received signal r_k , of length K_0 , then consists of s_k plus noise samples n_k :

$$r_k = s_k + n_k \quad k = 0, \dots, K_0 - 1$$

or using vector notation,

$$\mathbf{R} = \mathbf{S} + \mathbf{N}$$

where n_k are independent identically distributed Gaussian random variables with mean zero and variance σ^2 (i.e., $(N(0, \sigma^2))$).

From the received signal it is desired to estimate the transmitted signal, its period, and the noise intensity. Thus, the parameters to be estimated are $(q_0, \dots, q_{P-1})$, P , and σ^2 .

By considering $n_k = r_k - s_k$ to be a k -dimensional Gaussian random vector, the conditional probability density function of the received signal may be written as

$$p(\mathbf{R}/\mathbf{P}, \mathbf{Q}, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{K_0/2}} \exp \left(-\frac{1}{2\sigma^2} \sum_{k=0}^{K_0-1} (r_k - s_k)^2 \right). \quad (1)$$

The ML estimator maximizes (1), or equivalently maximizes

$$\ln p(\mathbf{R}/\mathbf{P}, \mathbf{Q}, \sigma^2) = -\frac{K_0}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{k=0}^{K_0-1} (r_k - s_k)^2 \quad (2)$$

as a function of $\mathbf{P}$, $\mathbf{Q}$, σ^2 at the received signal $\mathbf{R}$.

In view of the periodic nature of $\mathbf{S}$, the sum in the second term of (2) may be rewritten

$$\sum_{l=0}^{N-1} \sum_{k=0}^{P-1} (r_{k+lP} - q_k)^2 + \sum_{k=0}^{K_0-NP-1} (r_{k+NP} - q_k)^2 \quad (3)$$

where

$$N \triangleq \lfloor K_0/P \rfloor^1.$$

Note that the second term in (3) is absent when K_0 is an integer multiple of P . Thus

$$\frac{\partial \ln p(\mathbf{R}/\mathbf{P}, \mathbf{Q}, \sigma^2)}{\partial q_k} = \begin{cases} \frac{1}{\sigma^2} \sum_{l=0}^N (r_{k+lP} - q_k) & k < K_0 - NP \\ \frac{1}{\sigma^2} \sum_{l=0}^{N-1} (r_{k+lP} - q_k) & k \geq K_0 - NP. \end{cases}$$

Equating this to zero gives the ML signal estimate

¹ $\lfloor X \rfloor$ indicates the greatest integer less than X .

$$\hat{q}_k(P) = \begin{cases} \frac{1}{N+1} \sum_{l=0}^N r_{k+lP} & k < K_0 - NP \\ \frac{1}{N} \sum_{l=0}^{N-1} r_{k+lP} & k \geq K_0 - NP. \end{cases} \quad (4)$$

Likewise, setting $\partial/\partial\sigma^2 \ln p(\mathbf{R}/\mathbf{P}, \hat{\mathbf{Q}}, \sigma^2) = 0$ gives

$$\hat{\sigma}^2(P) = \frac{1}{K_0} \sum_{k=0}^{K_0-1} (r_k - \hat{s}_k)^2 \quad (5)$$

where

$$\hat{s}_k = \hat{q}_{k \bmod P}.$$

Thus it remains to choose $\hat{P}$ as that value of P which maximizes $\ln p(\mathbf{R}/\mathbf{P}, \hat{\mathbf{Q}}, \hat{\sigma}^2)$. Combining (4) and (5) with (2) shows that the maximum is achieved by minimizing

$$\sum_{k=0}^{K_0-1} (r_k - \hat{s}_k)^2$$

or, equivalently, by minimizing $\hat{\sigma}^2(P)$.

Combining (4) with (5) gives (see Appendix A)

$$\begin{aligned} \hat{\sigma}^2(P) &= \frac{1}{K_0} \left[\sum_{k=0}^{K_0-1} r_k^2 - \sum_{k=0}^{K_0-1} \hat{s}_k^2 \right] \\ &= \frac{1}{K_0} [\phi_R(0) - \mathcal{E}_{\hat{s}}(P)] \end{aligned} \quad (6)$$

where

$$\phi_R(k) = \sum_{j=0}^{K_0-1-k} r_j r_{j+k} \quad (7)$$

is the autocorrelation function of $\mathbf{R}$ and

$$\mathcal{E}_{\hat{s}}(P) = \sum_{k=0}^{K_0-1} \hat{s}_k^2 = (N+1) \sum_{k=0}^{K_0-NP-1} \hat{q}_k^2 + N \sum_{k=K_0-NP}^{P-1} \hat{q}_k^2 \quad (8)$$

is the energy of $\hat{\mathbf{s}}$. Thus $\hat{P}$ is that value of P which maximizes $\mathcal{E}_{\hat{s}}(P)$, and the problem is reduced to a one-dimensional maximization over P .

In the case where the observation is an exact integer number of periods, i.e., $K_0 = NP$, (8) becomes

$$\mathcal{E}_{\hat{s}}(P) = N \sum_{k=0}^{P-1} \hat{q}_k^2. \quad (9)$$

Applying (4) and (7) to (9) gives (see Appendix B)

$$\mathcal{E}_{\hat{s}}(P) = \frac{P}{K_0} \left[\phi_R(0) + 2 \sum_{l=1}^{N-1} \phi_R(lP) \right]. \quad (10)$$

Thus for K_0/P an integer $\mathcal{E}_{\hat{s}}(P)$ may be expressed either in terms of the signal estimate $\hat{q}_k(P)$ (8) or in terms of the autocorrelation of the received signal $\phi_R(k)$ (10). If N is sufficiently large, the value of $\mathcal{E}_{\hat{s}}(P)$ given by (10) is approximately that of (8), even for values of P which are not integer submultiples of K_0 . This expression in terms of the autocorrelation

function is useful both in possible implementation and in interpreting the frequency domain behavior of the ML estimator.

III. APPLICATION TO SPEECH SIGNALS

The ML period estimator $\hat{P}$ as found in the last section is that value of P which maximizes $\mathcal{E}_s(P)$, or equivalently minimizes $\hat{\sigma}^2(P)$. Examination of (5) shows that, in the absence of noise, the estimator $\hat{\sigma}^2(P)$ has zeros at $P, 2P, \dots$, and additionally at K_0 . With noise present, the zero at K_0 is the global minimum of $\hat{\sigma}^2(P)$, and the local minimum at $(l+1)P$ is less than that at $lP, l = 1, 2, \dots$. Thus the estimator derived in Section II must be modified for application to speech signals.

Examination of the estimator $\hat{\sigma}^2$ shows that it is biased. Caprio [4] has shown, for the case when the observation consists of an integer number of periods, $E(\hat{\sigma}^2/P) = ((N-1)/N)\sigma^2$. This suggests $\hat{\sigma}_{UB}^2 = (N/(N-1))\hat{\sigma}^2$ as an unbiased estimator, giving

$$\mathcal{E}_s'(P) = \frac{2P}{K_0 - P} \sum_{l=1}^{N-1} \phi_R(lP) \quad (11)$$

as the corresponding estimate of signal energy. Note that $\phi_R(0)$ has been omitted from the summation in (11). This results in the peaks of $\mathcal{E}_s'$ at $P, 2P, \dots$ having equal amplitude, whether or not noise is present.

In order that a signal of period P not be identified as having period $lP, l = 2, 3, \dots$, it is desirable to enhance the peaks at short periods at the expense of those at longer periods. This may be done by changing the factor $2P/(K_0 - P)$ in (11) to $2P/K_0$ giving

$$g(P) \triangleq \frac{2P}{K_0} \sum_{l=1}^{N-1} \phi_R(lP) \quad (12a)$$

$$= \mathcal{E}_s(P) - \frac{P}{K_0} \phi_R(0). \quad (12b)$$

The maximizer of $g(P)$ has been found to be the most satisfactory estimate of pitch period for voiced speech sounds. In particular, this weighting results in a suppression of the peaks at multiples of the pitch period without excessively increasing the heights of peaks due to strong harmonics.

Improved Resolution

By considering values of P which are not necessarily integers, $g(P)$ may be made to yield resolutions finer than one sampling period. Ideally a band-limited interpolation scheme should be used to provide the required values of ϕ_R , as Markel has done in his simplified invoice filtering technique (SIFT) algorithm [5], but simple zero-order spline interpolation (i.e., $\phi^*(lP) = \phi([lP])$) provides usable results.

For example, consider a signal of period corresponding to $3\frac{1}{2}$ samples with autocorrelation as shown in Fig. 1. Simply picking the first peak will give a period estimate of 3 whereas by forming $g(P)$ for $P = \frac{1}{2}, 1, 1\frac{1}{2}, \dots$ using interpolated values of ϕ_R gives a single peak at $3\frac{1}{2}$.

This increased resolution is particularly beneficial in the usually troublesome case of high-pitched speech [6], as will be shown in Section IV.

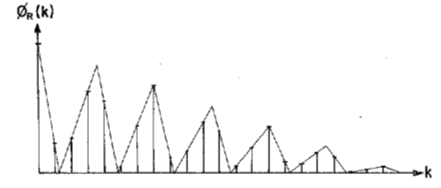


Fig. 1. Sampled autocorrelation of signal with period $3\frac{1}{2}$.

Voiced-Unvoiced Decision

In voiced speech the energy due to the periodic portion of the signal, $\mathcal{E}_s'(P)$, constitutes a large portion of the total signal energy, $\phi_R(0)$, whereas in unvoiced speech, the proportionate value of $\mathcal{E}_s'(P)$ would be small. Thus $\mathcal{E}_s'(P)/\phi_R(0)$ should be a useful parameter in a voiced-unvoiced decision.

Frequency-Domain Characteristics

It is instructive to consider the procedure in the frequency domain. $\mathcal{E}_s'(P)$ (or $g(P)$) may be interpreted as the inner product of a comb $C_P(k)$, of intertooth spacing P , with the autocorrelation of the received signal. This corresponds to the inner product of the transform of C_P with the power spectrum of R . In the case where K_0/P is an integer, the transform of C_P is also a comb, with spacing f_s/P where f_s is the sampling frequency (i.e., $\mathcal{F}[C_P(k)] = C_{f_s/P}(f)$). When K_0/P is not an integer, $\mathcal{F}[C_P(k)]$ is still comb-like with spacing f_s/P . Fig. 2 shows an example of this case for $K_0 = 256, P = 30$. Maximizing $g(P)$ is equivalent to finding the comb filter which passes the maximum signal energy. This compares with a previously proposed comb technique [7], [8] which seeks to null, rather than pass, the periodic component of the signal, and which operates on the signal itself, rather than its autocorrelation.

Another procedure related to this has been proposed by Steiglitz *et al.* [9], who use trigonometric curve fitting to minimize a least-square error similar to (5).

IV. IMPLEMENTATION

The pitch detection scheme as developed in the preceding section consists of forming $g(P)$ for various values of P and choosing $\hat{P}$ to maximize $g(P)$. $g(P)$ may be evaluated either from the autocorrelation function using (12a) (Method A) or from (12b) by forming $\mathcal{E}_s(P)$ using the signal estimator and subtracting $(P/K_0)\phi_R(0)$ (Method B).

If $g(P)$ is evaluated for $P = 0, 1, \dots, K_0 - 1$, the same amount of computation ($K_0^2/2$ multiplies, K_0^2 adds, and K_0^2 add-index's) is required for both methods assuming ϕ_R is evaluated directly [using (7)]. If K_0 is several times the maximum pitch period to be encountered, it is only necessary to evaluate $g(P)$ up to some $P_{\max} < K_0$. In this case the computation required for Method B is $P_{\max}^2/2$ multiplies, $K_0 P_{\max}$ adds, and $K_0 P_{\max}$ add-index's.

If a tracking algorithm is used, further limiting the range of P to be considered, a further saving may be obtained. On the other hand, if ϕ_R is computed using the fast Fourier transform (FFT), Method A offers a computational saving when $g(P)$ is required over a large range of P .

This scheme was implemented on a Texas Instruments 980B computer using a mixture of Fortran and assembly language

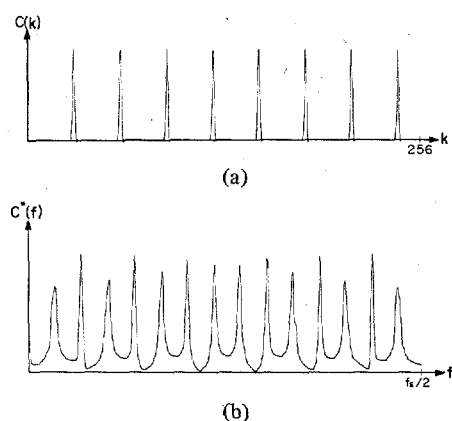


Fig. 2. (a) Time-domain comb with spacing of 30 and (b) corresponding frequency-domain comb.

programming. Utterances from several speakers were recorded on analog tape, low-pass filtered, digitized at a sampling rate of 8450 Hz and stored on disk. The resulting record was analyzed as a sequence of overlapping frames of length 384 ($= K_0$) with an increment of 128 samples between successive frames. For each frame $g(P)$ was evaluated using Method A and the maximizer $\hat{P}$ chosen as the pitch period.

Figs. 3 and 4 show the results from typical frames of speech. Fig. 3 is from a male speaker and Fig. 4 is from a female speaker. In each figure, (a) is a plot of the frame of speech samples being analyzed, (b) is the autocorrelation function of this record, (c) is the power spectrum, and (d) is $g(P)$ as given by (12a) calculated at integer values of P . Fig. 4 shows the advantage of the increased resolution afforded by noninteger P . In Fig. 4(d), where $g(P)$ has been evaluated at integer increments, the correct peak at $P = 40$ has been flattened and the peak at $P = 80$ is larger. In Fig. 4(e) $g(P)$ has been evaluated at increments of $\frac{1}{2}$ and the peak at $P = 40$ is largest.

Fig. 5(a) shows a plot of $\hat{P}$ versus frame number for the record in which the segment of Fig. 4 occurred. Fig. 5(b) is $\mathcal{E}_s(\hat{P})/\phi_R(0)$, which for this record is a good voiced-unvoiced indicator.

Performance in the presence of noise was examined by mixing a known amount of noise with the speech signal and processing the resulting record. Two types of noise were used: 1) synthetically generated white Gaussian noise and 2) the sound from a cooling fan, processed and recorded as were the speech signals. The latter is typical of environmental noise encountered in realistic situations and contains both a random component from the turbulence and a periodic component from the motor.

To form a basis for the evaluation, reference pitch contours were obtained by manual examination of the speech signal. These were compared with the values for $\hat{P}$ produced by the pitch detector over the intervals where the original speech was voiced. It is assumed that in an actual application the voiced-unvoiced decision will not be based solely on information provided by the pitch detector [10].

Two measures of performance were derived from this comparison: 1) percentage of gross errors and 2) the rms value of the pitch error over those voiced frames in which gross errors did not occur. The percentage of gross errors is defined some-

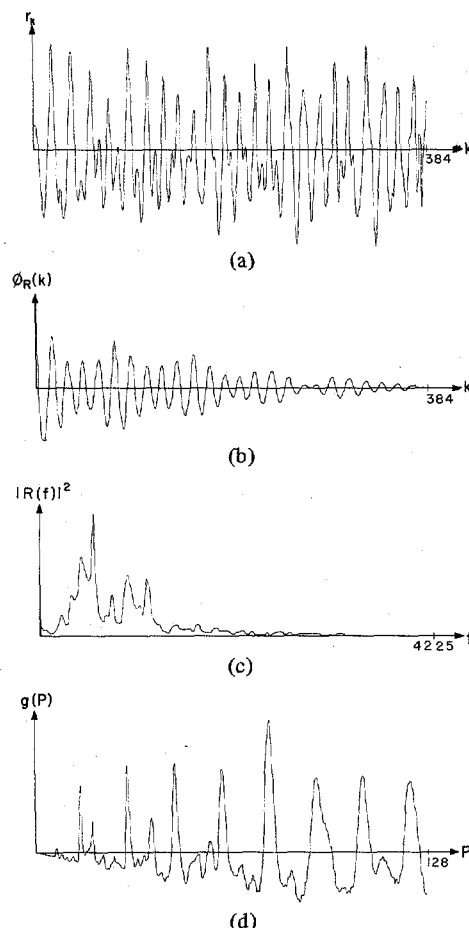


Fig. 3. (a) Speech signal, (b) autocorrelation, (c) spectrum, and (d) $g(P)$.

what arbitrarily as the ratio of the number of frames in which $|P - \hat{P}| > 5$ samples to the total number of voiced frames. This threshold corresponds, typically, to a 10 percent error.

No quantitative analysis was made of the gross errors encountered. However, the following qualitative observations were made: the errors in the cepstrum method tended to be random, due to the "background noise" of the cepstrum rising above the pitch peak; the errors in the ML method were generally either halving or doubling errors due to accentuation of a harmonic or subharmonic at the expense of the fundamental. As mentioned above, the viability of the procedure as a voiced-unvoiced decision is not being assessed.

Table I shows the performance of the ML pitch detection scheme for one female and two different male speakers with various types and levels of noise. For comparison, the performance of the cepstrum method [11], [12] is also given. For each of the three utterances analyzed the table gives the type of noise added, if any, the ratio of signal power to noise power, and the values of the two performance criteria achieved by both the ML and cepstrum methods.

V. CONCLUSIONS

The proposed pitch detection scheme based on an ML formulation has been shown to provide good behavior in ambient noise and to give increased frequency resolution. This latter property is advantageous in the normally difficult processing

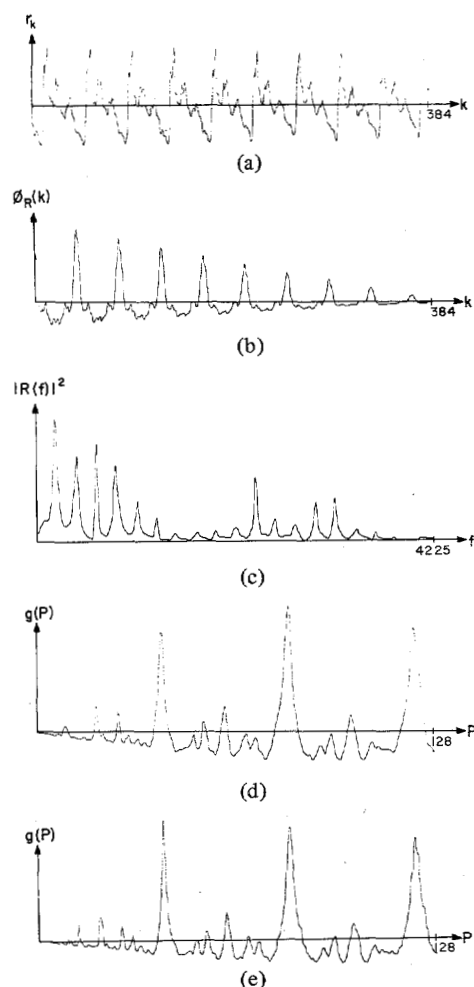


Fig. 4. (a) Speech signal, (b) autocorrelation, (c) spectrum, (d) $g(P)$ evaluated at $1, 2, 3, \dots$, and (e) $g(P)$ evaluated at $\frac{1}{2}, 1, 1\frac{1}{2}, \dots$.

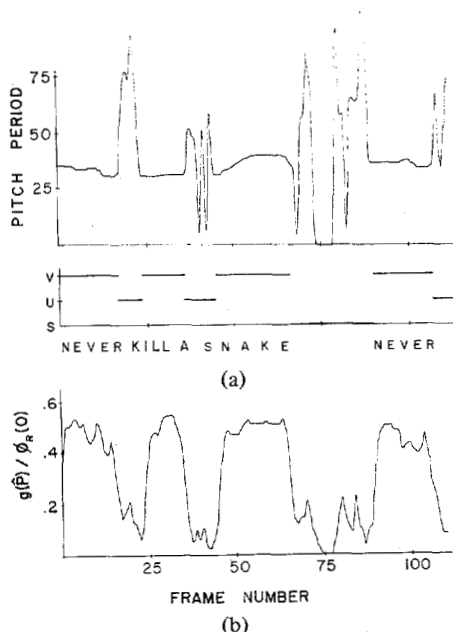


Fig. 5. (a) and (b) Pitch and $g(P)/\phi_R(0)$ versus frame number for a typical utterance. Voiced (V), unvoiced (U), and silent (S) sections are indicated.

TABLE I

Utterance*	Noise Type	SNR	% Gross Errors		RMS Error (in samples)	
			ML	CEPST	ML	CEPST
1	N	-	1.5	5.8	.70	.90
	W	4.0	4.4	13	.46	1.3
	W	2.0	2.9	13	.55	1.5
	F	4.0	13	20	.75	1.1
	F	2.0	16	19	.78	1.3
2	N	-	6.5	19	1.35	1.30
	W	2.0	18	43	.76	1.22
	W	1.0	18	50	1.01	1.21
	F	4.0	26	44	.71	1.23
	F	2.0	26	47	1.29	.97
3	N	-	14	32	1.23	1.37
	W	4.0	14	64	1.28	.92
	W	2.0	17	66	1.36	1.13
	F	4.0	34	58	.86	1.26
	F	2.0	41	61	.84	1.58

*Utterances:

1. Female: "Never Kill a Snake."
2. Male: "What Hath God Wrought?"
3. Male: "Beware the Ides of March."

Noise Types:

N. None

W. White Gaussian $N(0, \sigma^2)$

F. Fan Noise

$$\text{SNR} = \frac{\phi_S(0)}{\sigma^2}$$

of female speech (and may be useful in processing helium speech).

The availability of an accurate estimate for the pitch of a voiced-speech sound makes possible increased accuracy in determining its formant structure. In particular, the use of the signal estimator given by (4) in determining both poles and zeros of a vocal tract model is currently being studied.

The interpretation of the procedure as a comb filter suggests the possibility of detecting the periods of two distinct periodic signals summed together. This could form the starting point for a system capable of processing simultaneous multiple speakers.

APPENDIX A

Derivation of (6)

From (5)

$$\begin{aligned} \hat{\sigma}^2(P) &= \frac{1}{K_0} \sum_{k=0}^{K_0-1} (r_k - \hat{s}_k)^2 \\ &= \frac{1}{K_0} \left[\sum_{k=0}^{K_0-1} r_k^2 - 2 \sum_{k=0}^{K_0-1} r_k \hat{s}_k + \sum_{k=0}^{K_0-1} \hat{s}_k^2 \right]. \end{aligned} \quad (\text{A1})$$

The second summation of (A1) may be rewritten

$$\sum_{k=0}^{K_0-1} r_k \hat{s}_k = \sum_{j=0}^{K_0-NP-1} \sum_{l=0}^N r_{j+lP} \hat{s}_{j+lP} + \sum_{j=K_0-NP}^{P-1} \sum_{l=0}^{N-1} r_{j+lP} \hat{s}_{j+lP}.$$

Setting $\hat{s}_k = \hat{q}_k \bmod P$

$$\begin{aligned} &= \sum_{j=0}^{K_0-NP-1} \sum_{l=0}^N r_{j+lP} \hat{q}_j + \sum_{j=K_0-NP}^{P-1} \sum_{l=0}^{N-1} r_{j+lP} \hat{q}_j \\ &= \sum_{j=0}^{K_0-NP-1} \hat{q}_j \sum_{l=0}^N r_{j+lP} + \sum_{j=K_0-NP}^{P-1} \hat{q}_j \sum_{l=0}^{N-1} r_{j+lP}. \end{aligned}$$

From (4)

$$\sum_{l=0}^N r_{j+lP} = (N+1) \hat{q}_j \quad j = 0, \dots, K_0 - NP - 1$$

$$\sum_{l=0}^{N-1} r_{j+lP} = N \hat{q}_j \quad j = K_0 - NP, \dots, P - 1. \quad (A3)$$

Substituting (A3) into (A2):

$$\begin{aligned} \sum_{k=0}^{K_0-1} r_k \hat{s}_k &= (N+1) \sum_{j=0}^{K_0-NP-1} \hat{q}_j^2 + N \sum_{j=K_0-NP}^{P-1} \hat{q}_j^2 \\ &= \sum_{l=0}^N \sum_{j=0}^{K_0-NP-1} \hat{q}_j^2 + \sum_{l=0}^{N-1} \sum_{j=K_0-NP}^{P-1} \hat{q}_j^2 \\ &= \sum_{k=0}^{K_0-1} \hat{s}_k^2. \end{aligned} \quad (A4)$$

Substituting (A4) into (A1) gives

$$\hat{\sigma}^2(P) = \frac{1}{K_0} \left[\sum_{k=0}^{K_0-1} r_k^2 - \sum_{k=0}^{K_0-1} \hat{s}_k^2 \right]. \quad (6)$$

APPENDIX B

Derivation of (10)

Substituting (4) into (9):

$$\begin{aligned} \mathcal{E}_{\hat{s}}(P) &= N \sum_{k=0}^{P-1} \left(\frac{1}{N} \sum_{l=0}^{N-1} r_{k+lP} \right)^2 \\ &= \frac{1}{N} \sum_{k=0}^{P-1} \sum_{m=0}^{N-1} \sum_{\substack{|i-j|=m \\ 0 \leq j < i \leq N-1}} r_{k+iP} r_{k+jP} \\ &= \frac{1}{N} \sum_{k=0}^{P-1} \left[\sum_{l=0}^{N-1} r_{k+lP}^2 + 2 \sum_{m=1}^{N-1} \sum_{\substack{i-j=m \\ 0 \leq j < i \leq N-1}} r_{k+iP} r_{k+jP} \right] \end{aligned}$$

$$\begin{aligned} &= \frac{1}{N} \left[\sum_{l=0}^{N-1} \sum_{k=0}^{P-1} r_{k+lP}^2 + 2 \sum_{m=1}^{N-1} \sum_{\substack{i-j=m \\ 0 \leq j < i \leq N-1}} \sum_{k=0}^{P-1} r_{k+iP} r_{k+jP} \right]. \end{aligned}$$

Substituting $i = m + j$:

$$\begin{aligned} \mathcal{E}_{\hat{s}}(P) &= \frac{1}{N} \left[\sum_{j=0}^{K_0-1} r_j^2 + 2 \sum_{m=1}^{N-1} \sum_{j=0}^{N-M-1} \sum_{k=0}^{P-1} r_{k+(m+j)P} r_{k+jP} \right] \\ &= \frac{1}{N} \left[\sum_{j=0}^{K_0-1} r_j^2 + 2 \sum_{m=1}^{N-1} \sum_{n=0}^{K_0-1-MP} r_n r_{n+MP} \right]. \end{aligned}$$

Using (7) and the definition of N :

$$\mathcal{E}_{\hat{s}}(P) = \frac{P}{K_0} \left[\phi_R(0) + 2 \sum_{m=1}^{N-1} \phi_R(MP) \right].$$

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